

Ethan Tsao

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EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

B.S. in Statistics & Data Science, Technology Management Program

Dec. 2025

- **Coursework:** Regression Analysis, Statistical Machine Learning, Stochastic Process, Databases Design

EXPERIENCE

Software Engineer Intern

Jun. 2025 – Sep. 2025

Aquila Clouds

Pleasanton, CA

- Designed and implemented 3 interactive EChart visualizations tracking monthly cloud spend trends, resource utilization patterns, and cost anomalies across AWS/GCP platforms, enabling customers to identify 15-20% cost optimization.
- Developed Tableau dashboards analyzing cloud spending patterns across 50+ customer accounts, synthesizing billing data to surface key insights on cost drivers, usage trends, and budget variances, identifying potential savings opportunities.
- Reduced FinOps dashboard load times by 9% through refactoring 20+ React components and implementing data caching strategies, while conducting SQL-based analysis of cloud cost patterns to identify top spending drivers.

Alumni Affairs Intern

Oct. 2024 – Jun. 2025

Mosher Alumni Hall

Santa Barbara, CA

- Coordinated logistics for 60+ university events, managing vendor coordination, AV setup, and attendee experience.
- Maintained event tracking system in Excel documenting attendance patterns, expenditures, and resource allocation.

TECHNICAL PROJECTS

JobFlow and Market Intelligence Dashboard | *React, FastAPI, PostgreSQL, Python, Tableau*

Jan. 2026

- Built a full-stack job intelligence platform processing 80+ applications through a built-in Chrome extension; built Python ETL pipeline extracting 90+ technical skills and standardizing salary data with 81% parsing accuracy.
- Designed Tableau market analytics dashboard identifying compensation patterns: mid-sized firms (500-1K employees) pay 31% premium, hybrid work commands 17% salary advantage, and Snowflake expertise correlates with \$162K+ salaries.
- Developed 8 SQL queries analyzing role distribution, skills demand, and industry concentration, revealing Python (75%) and SQL (70%) as universal requirements while Software/Tech sector comprises 71% of data role opportunities.

Video Game Database | *MySQL, SQL*

Nov. 2025

- Architected a normalized relational database with 13+ tables using MySQL, implementing referential integrity constraints and validation rules to maintain data quality and enable efficient inventory management queries.
- Developed 15+ complex SQL queries utilizing multi-table JOINS, subqueries, aggregations, and window functions to extract business insights and support real-time reporting requirements for inventory analytics.

Bank Churn Prediction | *Python, Scikit-Learn, XGBoost*

Mar. 2025

- Developed predictive churn models (XGBoost, Random Forest, Bagging, K-NN, Logistic Regression) achieving 86.1% accuracy through stratified cross-validation and SMOTE balancing to address class imbalance.
- Conducted exploratory data analysis with statistical visualizations (density plots, correlation heatmaps, stacked bar charts) to identify age-based churn patterns and feature relationships across 10,000 customer records.
- Enhanced model performance through systematic preprocessing pipeline including min-max scaling, one-hot encoding, and IQR-based outlier removal, resulting in 0.852 ROC-AUC score with XGBoost classifier.

ORGANIZATIONS

Technical Consultant

Aug. 2025 – Oct. 2025

Wedge Health (YC S25)

Remote

- Conducted targeted outreach to healthcare organizations and provided consultative support for a health AI platform, educating potential stakeholders on AI-driven patient engagement solutions and identifying integration opportunities.

Project Member

Mar. 2025 – Sep. 2025

Gaucha Sports Analytics

Santa Barbara, CA

- Built Python web scraping pipeline extracting 3+ seasons of Big West Baseball statistics, performed ETL data cleaning, and created Tableau dashboards visualizing WAR/OBP metrics to inform lineup optimization for coaching staff.

TECHNICAL SKILLS

Languages & Core Analytics: Python (Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn), SQL (PostgreSQL, MySQL, SQL Server), R, Excel (Pivot Tables, Power Query)

Visualization & BI Tools: Power BI, Tableau, Looker, Plotly, ECharts

Big Data & Cloud: Azure, AWS, GCP, Git, Docker, JavaScript, React, FastAPI, REST APIs